Math 105L **Lecture 4-3**

The Derivative at a Point: Secant Lines to Tangent Lines

1. Review: Rates of Change & Graphical Meaning

- **Average Velocity / Rate of Change on $[a, a+h]$:**

$$\text{Avg RoC} = \frac{f(a+h)-f(a)}{(a+h)-a} = \frac{\mathbf{f(a+h)-f(a)}}{\mathbf{h}}$$

Graphical Meaning: The slope of the **secant line** connecting the two distinct points $(a, f(a))$ and $(a+h, f(a+h))$.

- **Instantaneous Rate of Change at $x = a$:**

$$\text{Inst RoC} = \lim_{\mathbf{h\to 0}} \frac{\mathbf{f(a+h)-f(a)}}{\mathbf{h}}$$

Graphical Meaning: The slope of the **tangent line** to the curve at the single point $(a, f(a))$.

Formal Definition of the Derivative at a Point Given a function $f(x)$, we define its **derivative at the point** $x = a$, denoted $f'(a)$, by:

$$\mathbf{f'(a)} = \lim_{\mathbf{h\to 0}} \frac{\mathbf{f(a+h)-f(a)}}{\mathbf{h}}$$

provided this limit exists. If the limit exists, we say f is **differentiable at** a.

2. Worked Classroom Example: Estimating $E'(7)$ for $E(t) = \frac{6}{t+1} + 4$

Let $E(t)$ represent student enrollment (in thousands) in a program t days after launch:

$$E(t) = \frac{6}{t+1} + 4$$

We estimate the instantaneous rate of change at $t = 7$ days using difference quotients:

$$\text{Avg RoC on } [7, 7+h] = \frac{E(7+h)-E(7)}{h} \quad \text{where } E(7) = \frac{6}{7+1} + 4 = \frac{6}{8} + 4 = 4.75$$

Sample Computation with $h = 0.1$:

$$E(7.1) = \frac{6}{7.1+1} + 4 = \frac{6}{8.1} + 4 \approx 4.74074$$

$$\text{Avg RoC} = \frac{4.74074 - 4.75}{0.1} = \frac{-0.00926}{0.1} = \mathbf{-0.0926} \text{ thousand students/day}$$

Behavior as $h \to 0$ from Both Sides:

h (days)	Avg RoC on $[7, 7+h]$	h (days)	Avg RoC on $[7, 7+h]$
-0.1	-0.0949	0.1	-0.0926
-0.01	-0.0939	0.01	-0.0936
-0.001	-0.0938	0.001	-0.0937

Conclusion:
As $h \to 0$ from both the left and right, the difference quotients converge to:

$$\mathbf{E'(7)} = \mathbf{-0.09375} \text{ thousand students/day} = \mathbf{-93.75} \text{ students/day}$$

The negative sign confirms that enrollment is dropping by approximately 94 students per day at $t = 7$.